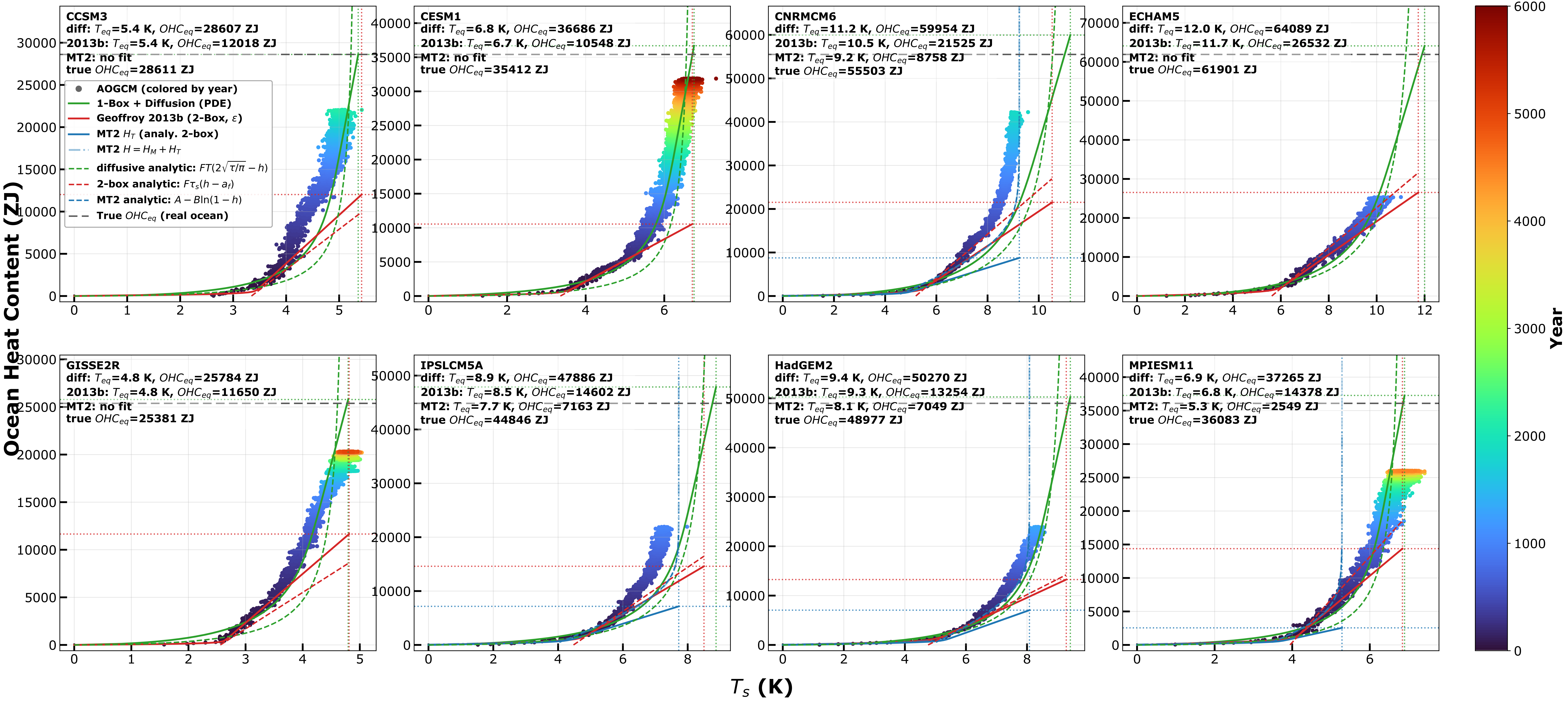
4xCO$_2$ OHC vs. Surface Temp: AOGCMs and three fitted models

02_forcing-50yravg parameters; MT2 H_T (analy. 2-box) from 01_replicate



CCSM3
diff: T_{eq}=5.4 K, OHC_{eq}=28607 ZJ
2013b: T_{eq}=5.4 K, OHC_{eq}=12018 ZJ
MT2: no fit
true OHC_{eq}=28611 ZJ
AOGCM (colored by year)
1-Box + Diffusion (PDE)
Geoffroy 2013b (2-Box, ε)
MT2 H_T (analy. 2-box)
MT2 $H = H_M + H_T$
diffusive analytic: $FT(2\sqrt{\tau/\pi} - h)$
2-box analytic: $F\tau_s(h - a_f)$
MT2 analytic: $A - B\ln(1 - h)$
True OHC_{eq} (real ocean)
CESM1
diff: T_{eq}=6.8 K, OHC_{eq}=36686 ZJ
2013b: T_{eq}=6.7 K, OHC_{eq}=10548 ZJ
MT2: no fit
true OHC_{eq}=35412 ZJ
CNRMCM6
diff: T_{eq}=11.2 K, OHC_{eq}=59954 ZJ
2013b: T_{eq}=10.5 K, OHC_{eq}=21525 ZJ
MT2: T_{eq}=9.2 K, OHC_{eq}=8758 ZJ
true OHC_{eq}=55503 ZJ
ECHAM5
diff: T_{eq}=12.0 K, OHC_{eq}=64089 ZJ
2013b: T_{eq}=11.7 K, OHC_{eq}=26532 ZJ
MT2: no fit
true OHC_{eq}=61901 ZJ
GISSE2R
diff: T_{eq}=4.8 K, OHC_{eq}=25784 ZJ
2013b: T_{eq}=4.8 K, OHC_{eq}=11650 ZJ
MT2: no fit
true OHC_{eq}=25381 ZJ
IPSLCM5A
diff: T_{eq}=8.9 K, OHC_{eq}=47886 ZJ
2013b: T_{eq}=8.5 K, OHC_{eq}=14602 ZJ
MT2: T_{eq}=7.7 K, OHC_{eq}=7163 ZJ
true OHC_{eq}=44846 ZJ
HadGEM2
diff: T_{eq}=9.4 K, OHC_{eq}=50270 ZJ
2013b: T_{eq}=9.3 K, OHC_{eq}=13254 ZJ
MT2: T_{eq}=8.1 K, OHC_{eq}=7049 ZJ
true OHC_{eq}=48977 ZJ
MPIESM11
diff: T_{eq}=6.9 K, OHC_{eq}=37265 ZJ
2013b: T_{eq}=6.8 K, OHC_{eq}=14378 ZJ
MT2: T_{eq}=5.3 K, OHC_{eq}=2549 ZJ
true OHC_{eq}=36083 ZJ
Ocean Heat Content (ZJ)
T_s (K)
Year
0
1000
2000
3000
4000
5000
6000